

It floats on water. It runs every **battery** you own.

Lithium is the lightest metal on Earth and the one the electric future is built on. The world mines just 240,000 tonnes a year, the price just crashed 80%, and most of it passes through one country. Here's the story.

8 kg/sec

The world mines about 8 kg of lithium every second.

Over a year that is only **240,000 tonnes**. We make more steel than that every **hour**. Lithium is proof that importance has nothing to do with how much you dig, and everything to do with what runs on it.

A little metal, doing a **huge job.**

EV batteries in a year

at ~8 kg of lithium per electric car

30 mn

Smartphones per EV

one car battery holds the lithium of thousands of phones

8,000

Density vs water

the only metal that floats, and it burns in water too

0.53

Light enough to float, reactive enough to power almost everything.

Crushed rock, or **evaporated brine.**

The same metal arrives two very different ways.

Hard rock: mine spodumene in Australia, crush it, roast it at 1,000 degrees. Fast to scale, but energy-hungry.

Brine: pump salty water from under South American salt flats into vast ponds and let the sun evaporate it for over a year. Far less carbon, but slow. One emits about five times the CO₂ of the other.

Australia digs it. China refines it.

Owning the mine is not the same as owning the supply chain.



The rock can come from anywhere. Turning it into a battery runs through China.

Up **tenfold**, then down **80%**.

Lithium is the most volatile major commodity of the decade.

As carmakers panicked about supply, the price spiked roughly **tenfold** into 2022. Then new mines flooded the market, EV growth cooled, and it **crashed about 80%**. The crash bankrupted projects that the next shortage will badly miss. Boom, bust, repeat.

It used to be glass. Now it's **batteries**.

A decade ago, lithium meant ceramics and lubricating greases. Today it is almost entirely about storing energy.

~90%

of demand is now batteries, mostly EVs

64%

was the battery share back in 2020

The clearest bet on the **electric future.**

Lithium is also the clearest example of how bumpy that road is. Demand follows EV sales, EV sales follow policy and price, and new mines take years to switch on, so supply lurches between glut and shortage.

Long term the direction is one way: more batteries, more grid storage, more lithium. The short term is whiplash.

Sources: USGS, IEA, SMM, Benchmark Minerals (2024 to 2025).

Next up: **nickel**, the other half of the battery.

I'm a materials engineer turned data analyst, breaking down one material at a time, the numbers, the science, and why it matters. Steel, aluminium, copper and rare earths came first. Save this, and follow for nickel.

Interactive dashboard + data on GitHub

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